BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, HYDERABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2022

CHEMISTRY

Past Paper 2022

Retyped faithfully from the original board paper — same questions, no changes.

Section-A Multiple Choice Questions (MCQs)

Note: Q.1 Choose the correct answer for each from the given options:

Q1. The formula of chile salt peter is:

(a) NaNO3
(b) NH4NO3
(c) KNO3
(d) AgNO3

Q2. Which one is secondary Alkylhalide:

(a) CH3 − CH2 − CH2 − Cl
(b) CH3 − C − Cl (CH3 above, H below)
(c) CH3 − C − Cl (CH3 above, CH3 below)
(d) None of these

Q3. The reaction in which single is converted in to double or triple bond is known as _____.

(a) Addition reaction
(b) Substitution reaction
(c) Elimination reaction
(d) Re-arrangement reaction

Q4. The process of lose of oxygen from a substance or addition of Hydrogen is called_____.

(a) Oxidation
(b) Reduction
(c) Hydrogenation
(d) None of these

Q5. Greater is the electro negativity of an element, stronger is the _____ character of its acid.

(a) Acid
(b) Basic
(c) Both (a) and (b)
(d) None of these

Q6. Urea was prepared in _____.

(a) 1828
(b) 1928
(c) 1840
(d) None of these

Q7. Aldehyde and Ketones we commonly referred to as _____.

(a) Carbonyl compound
(b) Penols
(c) Ethers
(d) None of these

Q8. In the electrolysis of NaCl _____ occurs at cathodes:

(a) Oxidation
(b) Reduction
(c) Redox reaction
(d) None of these

Q9. Which of the following element silicates are used in the manufacture of _____.

(a) Ca
(b) Cu
(c) Cr
(d) None of these

Q10. The mixture of sodium and calcium silicates are used in the manufacture of _____.

(a) Ordinary glass
(b) Pyrex glass
(c) Coloured glass
(d) All of these

Q11. Which one of following is mono saccharine?

(a) Maltase's
(b) Sucrose
(c) Gelatos
(d) Cellulose

Q12. Presence of double bond in a compound is the sign of _____.

(a) Saturation
(b) un-saturation
(c) substitution
(d) None of these

Q13. _____ forms polymeric hydrides.

(a) Na and Li
(b) B and Be
(c) Be and Mg
(d) Mg and Ca

Q14. Copper sulphate penta hydrated is also called:

(a) Green Viriol
(b) Black vitriol
(c) Blue vitiol
(d) All of these

Q15. Alkali metal are the element of _____

(a) Group IIIA
(b) Group IIA
(c) Grouup IA
(d) Group VIIIA

Q16. In Latin, the word petroleum means _____.

(a) Rock oil
(b) Cracked oil
(c) Rook oil
(d) All of these

Q17. IUPAC name of CH3−C(=O)−CH3 is:

(a) Di-methyl ketone
(b) Acetone
(c) 2- propa-none
(d) All of these

Q18. Third period of modern periodic table contain:

(a) 19 elements
(b) 2 elements
(c) 32 elements
(d) 8 elements

Q19. Natural gas contain methane gas (CH4) about _____

(a) 98.5%
(b) 94.60%
(c) 32 elements
(d) All of these

Q20. Which one is oxonium ion:

(a) R−O^{+}−R (H below O)
(b) $H3C^{+}$
(c) O−H
(d) None of these

Q21. De hydro halogenations is the reaction of:

(a) Addition of H2O
(b) Elimination of HX
(c) Addition of HX
(d) Elimination

Q22. Hydrides which have salt like character are known as _____.

(a) Metalic hydride
(b) Complex hydride
(c) Ionic hydride
(d) Polymeric hydride

Q23. Elements of sub group IIIA to VIIIA (Except He) are known as _____

(a) s-block
(b) p-block
(c) d-block
(d) f-block

Q24. Ketones are prepared by the oxidation of _____

(a) Primary Alchol
(b) Secondary Alchol
(c) Tertiary alcohol
(d) All of these

Q25. Benzene was first isolated by Mickle Faraday in _____

(a) 1845
(b) 1825
(c) 1945
(d) 1925

Q26. Tertiary carbon bonded directly to _____

(a) 2 hydrogens
(b) 3-coarbons
(c) 2 carbons
(d) 4 carbons

Q27. Hydrides of group IIIA & IVA are _____

(a) Neutral
(b) Acidic
(c) Basic
(d) None of these

Q28. Alkyle-halides undergoes _____

(a) Electrophilic substitution reaction
(b) Electrophilic and addition reaction
(c) Nucleophilic substitution reaction
(d) Nucleophilic addition reaction

Q29. _____ is the largest period in the periodic table.

(a) 4th period
(b) 5th period
(c) 6th period
(d) 7th period

Q30. Ruby is the oxide of _____

(a) Iron
(b) Copper
(c) Prophylene
(d) None of these

Q31. The systematic (IUPAC name of terbulyl Brremide is _____

(a) Tri Bromo Butane
(b) 1-Bromo Butane
(c) 1-bromo-1, 1dimethyl ethane
(d) 2-Bromo-2-methyl propane

Q32. Polyethene is the polymer of _____.

(a) Ethylene
(b) Acetylene
(c) Propyleen
(d) None of these

Q33. The vertical columns of element in the modern periodic table are called:

(a) Periods
(b) Blocks
(c) Groups
(d) Families

Q34. The color of chlorine (Cl2) gas is:

(a) Red
(b) Blue
(c) Yellow
(d) Greenish yellow

Q35. The valence shell of hydrogen is half filled like that of:

(a) Alkali metals
(b) Carbon family
(c) Halogen family
(d) Noble gases

Q36. Alkyl halide are classified into:

(a) 2 types
(b) 3 types
(c) 4 types
(d) 5 types

Q37. Formula of Iodoform is:

(a) CH3I
(b) CH2I2
(c) CHI3
(d) CI4

Q38. Mustard gas is prepared form:

(a) Ethene
(b) Ethyne
(c) Ethane
(d) Benzene

Q39. C11H2n is the general formula of:

(a) Alkane
(b) Alkyne
(c) ALkene
(d) All of these

Q40. Which of the following metal is called self protected out by:

(a) Fe
(b) Cu
(c) Al
(d) Ag

Q41. The industrial preparation of Nitric acid is carried out by:

(a) Contact process
(b) Cartner – Kelliner process
(c) Down's Process
(d) Ostward's method

Q42. The bond energy of Hydrogen molecule is:

(a) 102 K cal/mol
(b) 104 K.cal/mol
(c) 105 K.cal/mol
(d) 106 K.cal/mol

Q43. The first known organic compound synthesis in the laboratory is:

(a) Benzene
(b) Acetic acid
(c) Urea
(d) Ethane

Section-B (Short Questions)

> Note: Answer any SIX of the following:

Q2. Define any two of the following. Allotopy Elecrophile Enzymes

Q3. Differentiate SN1 & Sn2 rections.

Q4. Complete and balance the equations. (a) Al4C3 + H2O → Al2O3 + (b) PCl3 + H2O → ..... + HCl (c) MgN + H2O → Mg (OH)2 + ... (d) NaAlH4 + H2O → Al(OH)3 + ...

Q5. What are Alkylhalides ? How are they classified?

Q6. Give the reason: (a) Diamond is bad conductor of electricity. (b) Ionization potential decreases down the group. (c) Transition metals form coloured compounds. (d) PH3 is weaker base than NH3

Q7. What is Atomic hydrogen? How it is prepared?

Q8. Define and explain the Homologous series.

Q9. Write the structure of following: (a) m-nitrobenzoic acid (b) Ethly-acetate (c) Glycerol (d) Chloroform

Q10. Write a short note on corrosion.

Section-C Descitpive Answer

> Note: Answer any TWO of the following questions.

Q11. Explain Modern periodic Law. Describe various type of elements based on electronic configuration?

Q12. What are transition elements? Describe following properties of transition elements. (a) Atomic size (covalent radii) (b) Melting and boiling point

Q13. Write the IUPAC names of the following: 1. H2N − CH2 − CH2 −NH2 2. K2 [Cu9(CN)4] 3. [CO(NO2)3(NH3)3] 4. CH3 − CH − CH − CH3 (CH3, Br below) 5. CH2 − CH2 − CH − OH (OH, C2H5 below) 6. (CH3)2 − CH − C - C (CH3)3 (|| O below) 7. C2H6 − CH − CH2 − CH − CH (CH3, H3C, C2H5) 8. CH3 − CO − CO −C2H5 9. C2H5 − CH − CH2 − COOH (OH below)